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PALAEOMAGNETISM OF SOME LITHOLOGICAL UNITS FROM THE LOWER PALAEOZOIC OF THE VENDÉE, FRANCE

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ABSTRACT

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Results of the first palaeomagnetic study carried out in the west Vendean region at the southern edge of the Armorican Massif are presented. The lithological units sampled range from Middle to Upper Cambrian and Silurian. After alternating field demagnetization and thermal cleaning, La Lombardière basaltic rocks of Silurian age yielded only one residual component of magnetization ($D = 214°$, $I = +11°$) after tilt correction. The corresponding palaeopole is located at 28°S, 321°E. La Meilleraie sandstone and quartzite of Lower Ordovician age, which are equivalent to the Armorican sandstone, also contain one single residual component of magnetization ($D = 244°$, $I = +5°$) after demagnetization and tilt correction. The corresponding palaeopole is 15°S, 293°E. Finally, quartzites and rhyolites of Middle to Upper Cambrian age are characterized by two and possibly three components of magnetization. The mean direction of the main residual component after demagnetization and tilt correction is $D = 319°$, $I = +9°$; the corresponding palaeopole position is 40°N, 222°E.

The apparent path wander of the poles obtained in the Vendée is compared with the one of stable Europe (western) and Gondwana: a convergence of the poles from the Vendée and stable Europe is observed from the beginning of Ordovician time and a reasonable similarity at Silurian time. This study partly fills the gap of palaeomagnetic data for the period: Upper Cambrian–Silurian for the Armorican plate. As early as Upper Cambrian, the Vendée was an integral part if the Armorican plate. The plate tectonics scenario proposed in this paper is the following: accelerated separation of the Armorican block from Gondwana at Middle–Upper Cambrian time, motion of the Armorican block towards stable Europe at Ordovician time and proximity of Armorican plate to the northwest European craton at Silurian time. If this scenario is right, the nature of the Hercynian orogen is more likely the result of a collision, the Armorican plate acting as a buffer zone.

Palaeomagnetic studies of the Vendean Bocage rock sequence are considered important for future work.

INTRODUCTION

A data set of geological nature was used by Argand (1923) and Wilson (1966) to suggest the presence of an Early Palaeozoic proto-Atlantic ocean. Since then, many

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more or less complicated and plausible geological scenarios were put forward as an attempt to explain the presumed openings and closures of one or more ocean(s), relative continental motions and associated orogens (e.g., Cadomian, Caledonian and Hercynian). The nature of such orogens may be a collision, a subduction or a combination of the two, together with a number of regional variants. In particular, the Hercynian orogen of western Europe is characterized by a tectonic setting which is difficult to explain by either one of these two mechanisms. One approach to the study pertaining to the nature of this orogen consists of positioning the continental segments contained in this orogen (Morel and Irving, 1978, 1980, 1981) with the help of localized palaeomagnetic studies in order to obtain the pre-Hercynian configuration.

This approach may be partially or fully tested by comparing the Armorican Palaeozoic poles with poles of the same age originating from the stable European platform (Baltico-Russian and Central European) and the Gondwana supercontinent. Considering Palaeozoic studies of Vendean rocks from local areas with varying metamorphic grade and structural complexity should be helpful in determining to what extent the Hercynian tectonometamorphic events which took place between Lower Devonian ($\simeq$ 365 Ma) and Upper Carboniferous ($\simeq$ 280 Ma) times overprinted the previous magnetic memories inherent in the Vendean rocks. Another goal of this study is to consider whether the west Vendean region, located at the extreme southern part of the Armorican Massif, constitutes a separate unit (microplate) independent of the main Massif or whether it constitutes an integral part of the Armorican plate.

GENERAL GEOLOGY: STRATIGRAPHY, STRUCTURAL HISTORY AND METAMORPHISM

In the Armorican Massif, an important unconformity is located between the base of the Cambrian in Normandy and the units of Upper Precambrian age, a period extending from Lower Proterozoic (2000 Ma) to Upper Brioverian (600–650 Ma), which are affected by the Cadomian orogen. Of the four palaeogeographic domains identified by Cogné and Wright (1980) in the units constituting the basement of Variscan Europe, three are partially located in the present Armorican Massif (Fig. 1). In southern Brittany and the Vendée, it is quite difficult to differentiate the formations constituting the Brioverian series from the Lower Palaeozoic formations (Durand et al., 1977). Thus, there is a stratigraphic unconformity between Late Precambrian and Cambrian in the north Vendean region, but no equivalent well-defined unconformity in maritime Vendée.

Granitizations and Hercynian deformations of Devonian and Carboniferous age are superimposed to ante-Cambrian polymetamorphic and polyorogenic elements. The Armorican Massif remained emergent during the Palaeozoic transgressions (Matthews et al., 1980). During Mesozoic time, the limits of the Armorican Massif are defined by the extensions of the Jurassic and Cretaceous formations of the Paris

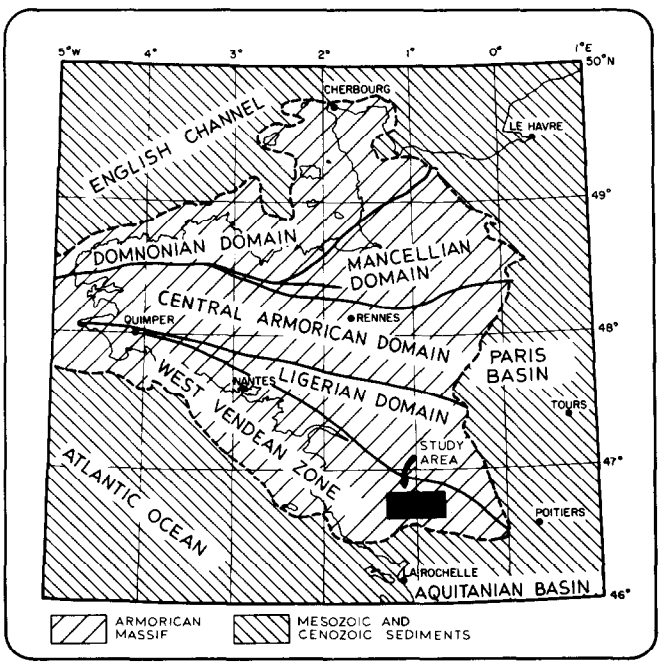


Fig. 1. Display of the main structural domains of the Armorican Massif (after J. Cogné, in Durand et al., 1977) and location of the area studied.

and Aquitaine basins. A few erosional remnants of Cenomanian overly the Palaeozoic basement. Hercynian shearing dynamics which is an integral part of the Variscan evolution in western Europe (Autran and Cogné, 1980) brought into contact the Precambrian and Palaeozoic compartments which probably were originally quite distant; they are now welded together by linear granitic masses intruded immediatly after the Hercynian orogen.

The Vendean area is characterized by two prominent NW–SE trending tectonic axes; these are (a) an axis of granitic rocks intruded into Precambrian terrane extending from Nantes to Parthenay during the Carboniferous, (b) Les Essarts-Mervent anticlinal axis composed of different metamorphic and folded Precambrian sequences (Gabilly et al., 1978). Between the biotite granitic massif of Pouzauges (Carboniferous) to the east and the Les Essarts-Mervent granulitic gneiss anticline (Brioverian) lies a zone (Chantonnay synclinorium, Fig. 2) occupied by volcano-sedimentary sequences of Lower Palaeozoic age (Wyns, 1979). From NE to SW, narrow and elongated synclines of Saint-Pierre-du-Chemin, La Châtaigneraie and Champdeniers-Puy-Hardy have been identified (Mathieu, 1972). The Precambrian is composed of the following units: amphibolites, amphibolitic porphyrite, garnetiferous amphibolitic psammite, misaschists, granulitic granite and eclogites (Ters, 1972;

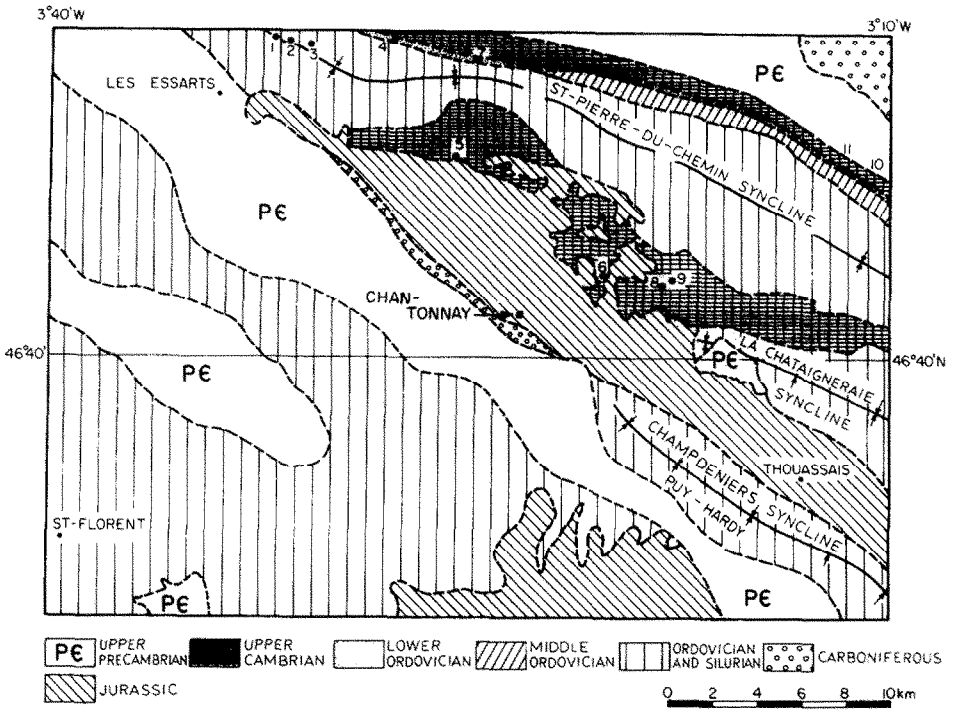


Fig. 2. Schematic geological map of the area described and location of the sampling sites. Geological map modified after the works of G. Mathieu (1972, 1976).

Mathieu, 1976). The Middle Cambrian marks the beginning of a transgression limited first to a few estuaries into which sandstones, quartzites, conglomerates and purple-coloured schists were deposited in the Chantonnay region. At Upper (?) Cambrian time, schistose rhyolites (ignimbrite-type) are widely distributed in the various synclines; these units suggest an important phase of acidic volcanism (Mathieu 1961, 1976). Some outcrops previously attributed to the Upper Cambrian by Mathieu (1976) would now appear to be of Ordovician age according to R. Wyns (pers. commun., 1982).

At Ordovician time, the transgression was definitely more widespread. The first marine deposits were essentially detrital. At the base, the Armorican sandstone is composed of sandstones, quartzites and arkoses with interbedded conglomerates and breccias (Mathieu, 1972, 1976). The upper sequence is composed of sericite schists, pyritic slates, ampelitic schists, phyllites, greywackes and phtanite bands; these units are occasionally slightly metamorphosed. The Silurian is represented by schists, sandstones, phtanites and nodular ampelites. This gritty and schistose sedimentation alternating with ampelites and phtanites is terminated by the outpouring of basic flows of tholeiitic composition (spilitic keratophyres) (C.G.F., 1980; Mathieu, 1976).

All these lithological units of Lower Palaeozoic age are metamorphored to the lower greenschist facies (sericite–chlorite).

SAMPLING AND TREATMENTS

The oriented samples collected in the field with the help of a magnetic compass were drilled and cut in the laboratory in order to obtain cylindrical specimens of 2.5 cm in diameter and 2.4 cm in length. Sites 1, 2 and 3 are representative of La Lombardière volcanic rocks of Silurian age. These basaltic rocks occur in the core of a Palaeozoic syncline in which the youngest sedimentary horizons are Silurian in age. Sites 7, 10 and 11 are quartzites and sandstones near La Meilleraie and north of Mouchamps; they have an Arenigian age (Lower Ordovician) and are equivalent to the Armorican sandstone of Britanny. Sites 4, 5, 6, 8 and 9 are of Upper Cambrian age and are located at La Bonnière (NE of Saint-Vincent-Sterlange), W and ESE of Sigournais, and at La Garneraie; these Cambrian units consist of sandstones, quartzite (yellow and red), conglomerate (Sigournais) and rhyolites within the Saint-Pierre-du-Chemin syncline. In his doctoral thesis (3rd cycle, unpublished), R. Wyns has shown that the Sigournais Formation lies stratigraphically above the Armorican sandstone and grades laterally into the Réaumur Group (Ordovician); however, the lack of a characteristic Ordovician fauna precludes a confirmation of this age relation.

Natural remanent magnetization (NRM) measurements were done with the help of a Schonstedt DSM-1A spinner magnetometer coupled with microcomputers (PDP-11 and APPLE II +). Thermal cleanings were done in a non-magnetic oven (residual field = 8 nT) model TSD-1 manufactured by Schonstedt Ltd. In the case of alternating field demagnetizations, either a GSD-1 Schonstedt apparatus (max. intensity = 100 mT) or a demagnetizer built at Laval University (Seguin, 1975) including a triaxial tumbler at three mu-metal concentric cylinders were used. This last apparatus can carry out demagnetizations up to 80 mT (peak-to-peak). All demagnetized NRM data were plotted on Zijderveld diagrams (Zijderveld, 1967) and where needed, various components of magnetization were isolated by vector subtraction techniques.

RESULTS

La Lombardière Silurian basaltic rocks

In-situ NRM directions of Silurian volcanics are mainly SW and less commonly NW with positive inclinations ranging from shallow to intermediate (Table I, Fig. 3). Initial intensities vary from $7 \cdot 10^{-3}$ to $6 \cdot 10^{-2}$ Am^{-1}. Alternating field (AF) and thermal demagnetizations were carried out an all the metavolcanic samples. These demagnetization techniques generally yield the same directions but each technique

TABLE I

Directions of natural remanent and residual magnetizations of the lithological units studied

Site No.	Component No.	Treatment	Polarity	n	In-situ		$\alpha_{95}(°)$	Tilt corrected			Strike/dip
					$D_0(°)$	$I_0(°)$		K	$D_1(°)$	$I_1(°)$	
Silurian volcanic rocks											
1		NRM	N	15	215	14	19.2	24	229	06	115/40
	1	(60–100 mT; 520°–580°C)	N	12	226	14	8.4	122	239	08	
2		NRM	N	12	224	45	30.5	11	220	10	110/85
	1	(30–60 mT; 550°–590°C)	M	11	222	49	23.8	15	218	27	
3		NRM	N	8	342	44	38.4	7.5	220	02	95/70
	1	(50 mT; 500°–590°C)	M	7	183	−14	25.0	22	189	11	
Lower Ordovician Armorican sandstone											
7		NRM	N	13	308	08	30.7	3	309	10	85/90
	1	(50–100 mT; 560°–610°C)	M	13	259	40	27.4	39	242	04	
10		NRM	M	9	268	70	59.5	4	242	03	
	1	(50 mT; 540°–600°C)	N	8	264	68	23.1	15	242	05	

11		NRM	N	10	288	29	58.9	3	274	05	
	1	(60–100 mT; 500°–590°C)	N	9	254	44	20.1	42	247	10	
Upper(?) Cambrian sandstone and rhyolite											
4		NRM	M	8	301	−05	40.0	4	311	06	115/40
	1	(50–90 mT; 500°–620°C)	R	7	311	−21	22.6	16	319	08	
5		NRM	N	14	288	−48	34.6	4	329	10	110/85
	1	(540°–610°C)	M	5	321	−39	25.5	18	329	27	
	2	(35–70 mT; (500°–570°C)	M	7	281	20	28.7	7	259	02	
8		NRM	N	10	307	−37	36.6	5	321	11	95/70
	1	(25–60 mT; 500°–570°C)	M	5	318	−53	23.1	14	339	10	
	2	(60–100 mT; 570°–620°C)	M	4	273	10	19.8	36	280	04	
9		NRM	M	13	265	13	20.8	19	252	03	85/90
	1	(20–70 mT;)	M	5	277	−64	16.2	13	331	05	
	2	(40–100 mT; 550°–620°C)	M	8	270	03	18.4	24	262	05	
	3	(60–100 mT; 570°–610°C)	N	5	222	75	25.3	14	186	10	

Legend: N = normal polarity; R = reverse polarity; M = mixed polarities; n = number of samples; D = declination; I = inclination; α_{95} = circle of confidence at 95% level; K = precision parameter.

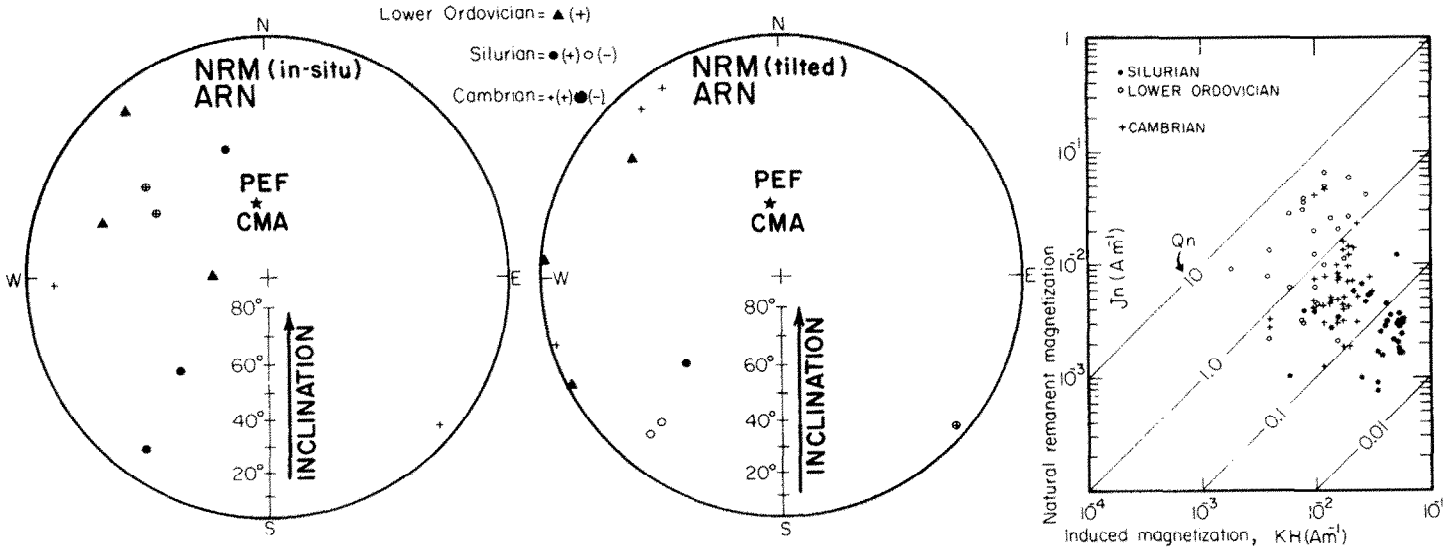


Fig. 3. Natural remanent magnetization directions shown for each site on a Wulff net (formations in-situ and after tilt correction). Bold (open) symbols represent the projections on lower (upper) hemisphere. The star shows the location of the present Earth's field at the sampling sites. NRM intensity (J_n in Am^{-1}) versus induced magnetic composent (KH in Am^{-1}) for individual samples and Koenigsberger's ratios (Q_n).

presents specific problems. For AF demagnetizations (Fig. 4), the median destructive field varies between 15 and 40 mT (mean 25 mT). An anhysteretic magnetization is noted from 80 mT upwards in 60% of the cases and at this stage the relative intensity of residual magnetization varies between 5 and 20% of the initial NRM (mean: 15%). In 10% of the cases, there are coercivities which exceed 100 mT. For these two reasons, it appears clearly that thermal cleaning (TC) is more effective than AF demagnetization and 75% of the specimens were thermally cleaned (Fig. 5).

Of the 25 samples which underwent systematic TC (100°, 200°, 300°, 400°, 450°, 500°, 525°, 550°, 570°, 580°, 590°, 600°, 610° and 620°C), 20 exhibited an unblocking temperature (T_{UB}) ranging from 500° to 575°C (mean 550°C) and two others from the same site slightly lower T_{UB} (mean 510°C). One sample (2.3) has no T_{UB}; its intensity increases from 450°C upwards. Two last samples (1.4 and 2.4) have T_{UB} at 600° and 590°C (sample 2.4 has an antiparallel direction). These observations indicate that the majority of magnetization is due to almost pure magnetite and this was confirmed by microscopic observations; the remaining magnetization is apparently located in specular hematite. At the T_{UB}, the relative intensity of residual magnetization varies between 2 and 15% (mean: 10%) of the initial NRM. In 65% of the cases, the intensity of residual magnetization increases in the temperature range: 575°–610°C (mean: 595°C) and this suggests remagnetizations caused by mineralogical changes (oxidation?, formation of hematite). Site 3 results required a more elaborate analysis because three of the samples are characterized by mixed polarities and one of these (3.2) certains a normal and a reversed direction, each corresponding to different mineralogies (T_{UB} at 500° and 600°C). The normal polarity corresponds to magnetite and the reversed to hematite.

The directions of residual magnetization (RM) after AF demagnetization and TC are generally concordant and shown to be univectorial (component No. 1, Figs. 4 and 5) in conformity with the Zijderveld diagrams (Zijderveld, 1967). A weak viscous component was detected in a few instances. The mean characteristic direction of the in-situ RM (Table II) is $D = 209°$, $I = 17°$. After tilt correction, the mean direction is $D = 214°$, $I = 11°$. The K_b/K_a ratio which is the ratio of precision parameters (Fisher, 1953) after and before tilt correction of the mean sample directions suggests a fold test which is slightly positive even though it is non-significant at the 95% confidence level (McElhinny, 1964). Assuming a primary pre-tilt magnetization for the La Lombardière Silurian volcanics, a palaeopole of 25°S, 321°E is obtained. If this is not the case, the mean characteristic magnetization of La Lombardière volcanics is close to that obtained for rocks of Hercynian age (Upper Carboniferous) in the Armorican Massif, that is $D = 200°$, $I = 9°$ for Ploumanac'h granite (Duff, 1979) and $D = 203°$, $I = 13°$ for Flamanville granite (Van der Voo and Klootwijk, 1972).

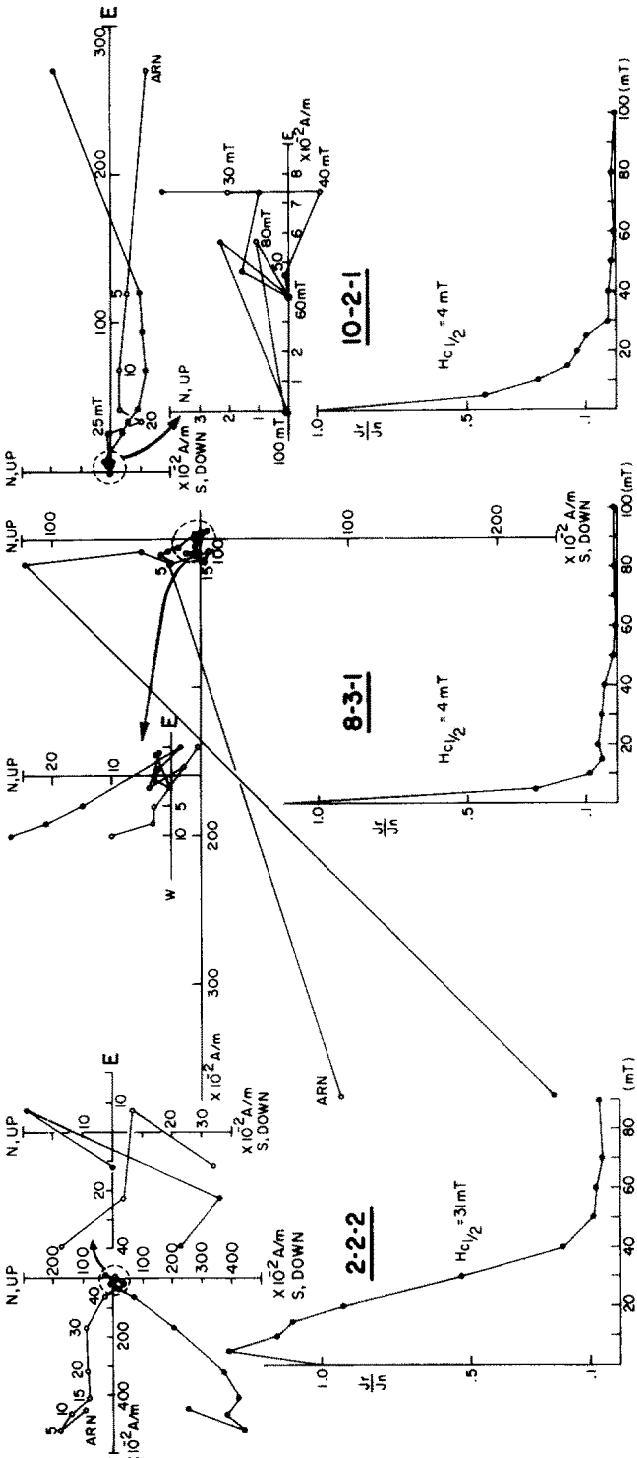


Fig. 4. Typical orthogonal diagrams (Zijderveld, 1967) for successive points of the magnetization vector for the three geological formations: Cambrian (8-3-1), Ordovician (10-2-1) and Silurian (2-2-2). Bold (open) circles represent projections in the horizontal (vertical) plane. The numbers close to the circles represent the demagnetization steps (in mT).

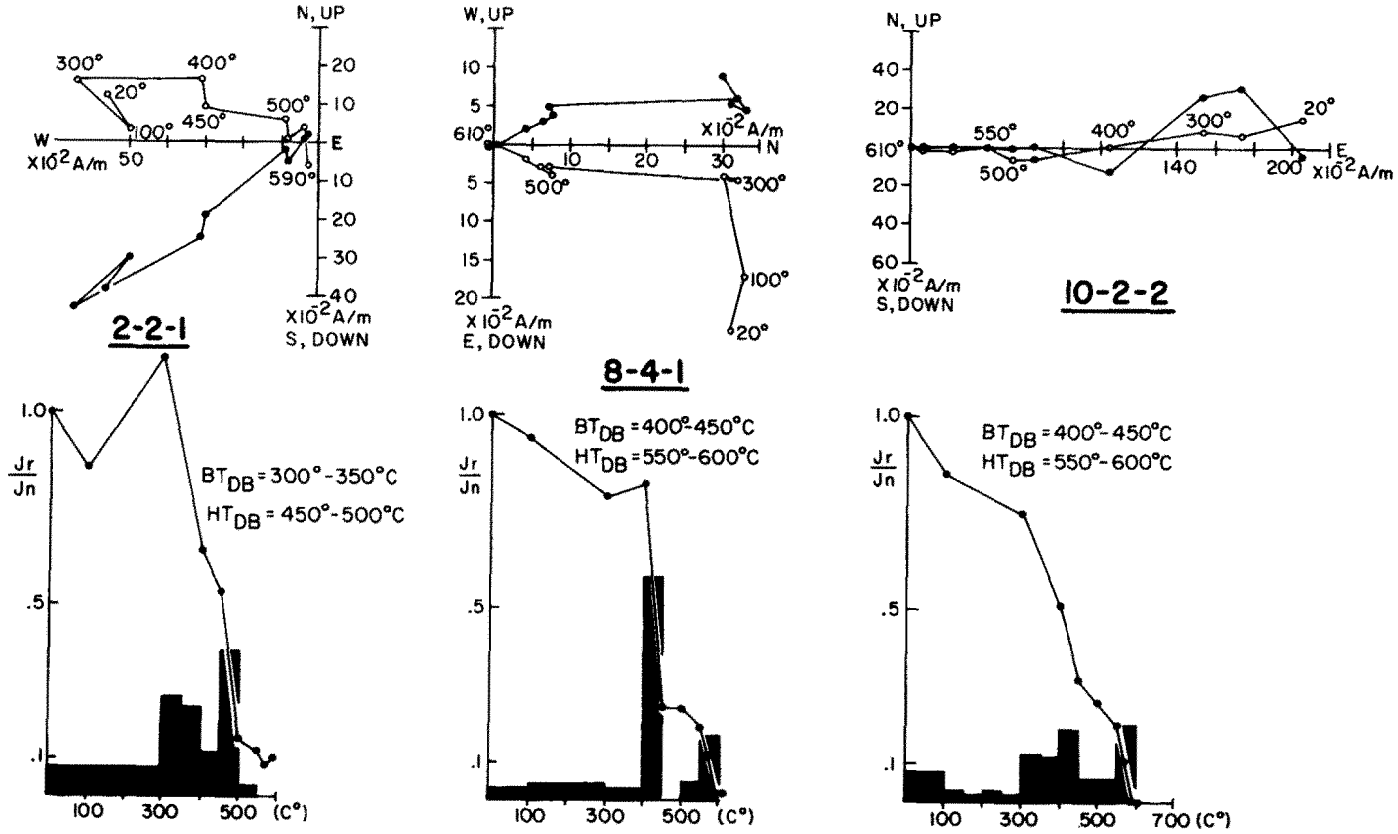


Fig. 5. Typical orthogonal diagrams (Zijderveld 1967) for successive points of the magnetization vector for the three geological formations: Cambrian (8-4-1), Ordovician (10-2-2) and Silurian (2-3-1). Bold (open) circles represent projections in the horizontal (vertical) plane. The numbers close to the circles represent the heating steps (in °C).

TABLE II

Mean directions of magnetization and palaeopoles of the lithological units studied

N	Component No.	In-situ		$\alpha_{95}(°)$	K_a	Tilt corrected		$\alpha_{95}(°)$	K_b	K_b/K_a
		$D_0(°)$	$I_0(°)$			$D_1(°)$	$I_1(°)$			
Silurian volcanic rocks										
3	1	209	17	29	5	214	11	22	7	1.4
Palaeopole						28°S	321°E			
Lower Ordovician Armorican sandstone										
3	1	258	51	23	28	244	05	14	80	2.9
Palaeopole						15°S	293°E			
Upper(?) Cambrian sandstone and rhyolite										
4	1	310	−45	25	14	329	09	19	24	1.7
Palaeopole						40°N	222°E			
3	2	274	11	16	64	267	01	19	44	0.7
Palaeopole		07°N	270°E							
1	3	222	74	–	–	186	−10	–	–	

Legend: D, I, α_{95}: same definition of the symbols as in Table I; K_a = in-situ precision parameter; K_b = tilt corrected precision parameter.

(b) La Meilleraie quartzites and sandstones (Armorican sandstone)

Only one RM component was detected in the La Meilleraie Ordovician Formation. As is the case for the La Lombardière volcanics, an anhysteritic magnetization is observed from 60 to 80 mT in 50% of the samples which were AF demagnetized; for half of the samples of site 7, some occasional coercivities are higher than 100 mT. Except for these samples having coercivities equal to or larger than 100 mT, the median destructive field ranges from 6 to 60 mT (mean: 20 mT) and the relative intensity of the RM varies between 5 and 40% of the initial NRM (mean 25%). Because AF demagnetization of 12 samples was shown to be inefficient and leading to uncertain results, a systematic TC (100°–630°C) was carried out on the remaining 20 samples.

In-situ NRM directions of La Meilleraie metasediments are located mainly to the west with positive inclinations ranging from shallow to steep (Table I) . The initial intensities range from $2 \cdot 10^{-3}$ to $2 \cdot 10^{-2}$ A_m^{-1}. Pilot AF demagnetizations (Fig. 4) indicated the presence of a magnetization due to both magnetite and hematite; it was thus necessary to carry out TC (Fig. 5). In 75% of the cases, the T_{UB} lies between 460° and 540°C (mean 520°C); this indicates that the mineral responsible for the magnetization is almost pure magnetite. In 20% of the cases, the presence of two T_{UB}s is noted, that is one between 450° and 540°C and another between 610° and 630°C (hematite or specularite). Sample 10.3 and half of the samples from site 7 are characterized by there two T_{UB}s. The other half of the site 7 samples registered only one T_{UB} ranging from 610° to 630°C (mean: 620°C). At the T_{UB} point, the normalized intensity of the RM varies between 2 and 25% (mean 10%) for all the samples having a T_{UB} at 540°C or else two T_{UB}s; in 30% of the cases, the RM intensity increases in the temperature interval: 580°–610°C. This increase is either caused by mineralogical changes or else by the fact that the memory in the magnetite disappears gradually leaving only that of hematite. Where the T_{UB} is at 620°C the relative RM intensity varies from 36 to 60% (mean: 45%); in only one sample (7.3) does the RM intensity increase above 620°C. But for a few rare exceptions, the RM directions after AF demagnetization or TC are concordant and the RM is generally univectorial (Figs. 4 and 5). The in-situ mean characteristic direction of the RM (Table II) is $D = 258°$, $I = 51°$. After tilt correction, the mean direction becomes $D = 244°$, $I = 05°$ and the K_b/K_a ratio suggests a fold test with a definite positive tendency, even though it is not significant at the 95% confidence level. Assuming a pre-folding primary origin for this RM, the corresponding paleopole is: 15°S, 293°E. Thus the mean characteristic direction of these La Meilleraie Ordovician rocks do not appear to coincide with that of magnetizations of Hercynian age within the Armorican Massif which implies that there is no question of an Hercynian overprint in these rocks.

(c) Upper (?) Cambrian quartzites and rhyolites

In-situ NRM directions of Cambrian volcano-sedimentary rocks are mainly located in the NW quadrant, and less commonly in a westerly direction and have shallow to intermediate but generally negative inclinations (Table I). The initial intensities vary from $9 \cdot 10^{-3}$ to $3 \cdot 10^{-2}$ Am^{-1}. The two demagnetization techniques (AF and TC) do not always yield the same results and both present particular problems. In the course of AF demagnetizations, the median destructive field varies between 5 and 25 mT (mean: 15 mT) with only one exception (metarhyolite sample No. 5.2, 60 mT). An anhysteritic magnetization (AM) takes place between 60 and 100 mT (mean: 80 mT) in 55% of the pilot samples. Before the introduction of the AM, the RM relative intensity is located in the 4–30% range (mean: 10%). No coercivities greater than 100 mT were observed. With the exception of site 4, various components of RM were extracted. The RM orientations after AF demagnetization and TC were reasonably concordant (Figs. 4 and 5), however an additional component having a southern direction ($D = 193°$, $I = 06°$) was isolated with AF demagnetization on site 5. This direction was found in only two samples (5.2.1 and 5.2.2) after TC with T_{UB}s at 610 and 565°C. Since TC was found to be more efficient than AF demagnetization for all sites with the possible exception of site 9, it was decided to filter out this southerly component.

Results obtained with TC are more significant than those isolated by AF demagnetization. This is why it was decided to use 60% of the pilot samples for TC. Twenty three (23) of these samples have a T_{UB} ranging from 450° to 550°C (mean: 525°C); two (5.2 and 8.1) are located between 580° and 620°C and three (4.1, 5.1 and 5.8) are characterized by the occurrence of two T_{UB}s, one at 560°C and another (variable) located between 590° and 620°C. The minerals responsible for the magnetization are fine-grained magnetite and specularite. At the T_{UB}, the RM relative intensity varies without exception between 1 and 15% (mean: 8%). In 50% of the cases, the RM intensity increases in the interval: 580°–620°C. This implies remagnetization caused by oxidation of magnetite or else that most of the magnetization of magnetite has been erased and only that of specularite remains.

Only one component (reverse polarity) was isolated on site 4. This same component (No. 1) was isolated on sites: 5, 8 and 9 (Table I). In-situ mean direction is $D = 310°$, $I = 45°$ and it is $D = 329°$, $I = 09°$ after tilt correction; the K_b/K_a ratio indicates a slightly positive fold test, but it is not significant at the 95% confidence level. Assuming a pre-folding primary component, the corresponding palaeopole position is 40°N, 222°E. This NW component is characterized by mixed polarities at three of the four sites, occasionally high coercivities (60 mT at site 5 and 70 mT at site 9) and relatively high T_{UB}s (600°, 590°, 580° and 590°C at sites 4, 5, 8 and 9 respectively). These characteristics signify the presence of hematite. Component No. 2 (western direction) is found on sites 5, 8 and 9 (Table I) and is characterized by mixed polarities and relatively high T_{UB}s. The component is observed in both lower

(530°–570°C) and higher T_{UB}s (750°–620°C). This suggests that the component is recorded in magnetite as well as specular hematite. RM directions after AF demagnetization and TC are concordant most of the time. In-situ NRM mean characteristic direction (Table II) is $D = 274°$, $I = 11°$ with a K_b/K_a ratio of 0.7; this suggests a negative fold test though its significance does not attain the 95% confidence level. The in-situ and tilt corrected directions of component No: 2 in these rocks are not very different, when considering the α_{95} values, from the tilt-corrected directions determined for the Lower Ordovician La Meilleraie Formation. The corresponding palaeopole position is 07°N, 270°E. A third component (southern orientation) was isolated on site 9 and on a few samples of site 5. This component is rather uncommon and is mainly isolated with AF demagnetization. The few TC specimens which yielded this component have a T_{UB} equal to 570°C. Given the small participation of this component to the whole set of data within this study, it appears unwise to insist on this particular component and to calculate its palaeopole position.

In summary, Silurian volcanic rocks and Lower Ordovician sandstones from the Vendée are characterized by a single component of magnetization while Upper(?) Cambrian sandstones and rhyolites recorded two (and possibly three) components of magnetization. Inasmuch as no event of geological importance apparently took place between Cambrian and Ordovician in the Armorican Massif, it seems more logical to expect the same number of components within units for these two geological periods. However, other researchers have also found two or three components of magnetization in Cambrian lithological units of the Armorican Massif. For instance, Hagstrum et al. (1980) identified two characteristic components of magnetization for the Keralain gabbro and Lézardrieux ignimbrites in Brittany and Duff (1980) isolated three successive phases of magnetization in some volcanic rocks and dykes and also in granitic intrusives (Duff, 1978) on Jersey Island. Even if the existence of multiple components is certain, the magnetization directions are quite variable from one palaeomagnetic study to the next.

Most of the palaeomagnetic studies undertaken in the Armorican Massif concern rocks of Lower to Middle Cambrian age, that is over a rather large time interval. Considering the fact that the α_{95} angles are generally large and that the time interval involved is large, the mean palaeopoles obtained for this period ought to be considered with some misthrust. Only one palaeomagnetic study of Upper Cambrian lithological units is available in the Armorican Massif for comparison with the results of the present study. Jones et al. (1979) obtained a mean direction of 312°, −01° on the Paimpol–Bréhec redbeds of Brittany; this direction is comparable to be value of 329°, 09° obtained in this study, taken into account the fact that the α_{95} angles are large. The results of this study, though they be of a preliminary nature, would thus appear to match those of Jones et al. (1979) and it is therefore considered important to carry out additional studies in this region.

APPARENT POLAR WANDER (APW) PATH OF THE LOWER PALAEOZOIC FOR WESTERN EUROPE, GONDWANA AND THE ARMORICAN BLOCK

Any attempt to situate the palaeopoles of a restricted block such as that the Armorican Massif with respect to the palaeopoles characteristic of the adjoining larger plates represented by western Europe and Gondwana is related to the more general problem of plate tectonics which incorporates the relationships between older geological blocks within a more recent and widespread orogenic event. Palaeomagnetic studies carried out within deformed terranes of a block size such as that of this study are necessarily characterized by a larger uncertainty in the interpretation of the results.

Inasmuch as the palaeomagnetic data are sparse for any valid synthesis of this problem, I have made an attempt to minimize this constraint by choosing specific formations for different blocks or terranes which cover a large time interval relative to that of the Hercynian orogenic event. Other approaches are possible, but all of these are subject to uncertainty. One of the first points to establish is the spatial relationships between the Armorican Massif and the surrounding plates, that is stable Europe and African Gondwana, during the Upper Cambrian to Silurian period. The APW path of Lower Palaeozoic poles of western Europe is poorly defined. The APW paths published to date are based mainly on results from Great Britain, Scandinavia and to a lesser degree from France. The quality of the palaeomagnetic data available from this part of the globe is often inadequate (see, e.g., Briden and Duff, 1981, and Van der Voo et al., 1980). Cambrian, Ordovician and Silurian palaeopoles of acceptable quality for stable Europe are shown in Table III. The mean of the palaeopoles obtained for the Cambrian period is 251°E, 52°N. At Upper Cambrian time (Fig. 6), Hagstrum et al. (1980) place the pole for stable Europe at approximately 15°E, 20°N, while Briden and Duff (1981) position it at 345°E, 25°N. For this some period, Hagstrum et al. (1980) position the pole of the Gondwana supercontinent at 0°, 25°N making use of some 6 palaeomagnetic studies while Brock (1981) puts it at approximately 290°E, 5°N, using another set (5) of palaeomagnetic studies. Making use of another set of palaeomagnetic data together with evidences of continental glaciation, Crowell (1981) sets the Upper Cambrian pole at 35°E, 50°N. Thus, for the Upper Cambrian period, the palaeopole obtained for the Vendée does not correspond to the ones determined for either stable Europe ($\simeq$ 330°E, 30°N) or Gondwana.

To the extent that the magnetization is primary one must conclude that the Armorican block was independent of the two other continental plates at Upper Cambrian time. However, it appears that the Vendée was then part of the Armorican block since the palaeopole (243°E, 50°N) obtained by Jones et al. (1979) in Brittany is not significantly different from the one obtained in the Vendée (222°E, 40°N). Alternatively, if the magnetization isolated is secondary, the palaeopole corresponding to the mean direction of magnetization (31°, −45°) is incompatible with those

of stable Europe and Gondwana for Devonian and Carboniferous times.

Briden and Duff (1981) positioned the pole of stable Europe during Ordovician time (Fig. 6) at approximately 10°E, 10°S. For the same period, Brock (1981) placed the Gondwana pole at approximately 340°E, 25°N and Crowell (1981) at 10°E, 20°N. Consequently, the pole of the Armorican block at Ordovician time (approximately 300°E, 15°S) lies close to the one of stable northwestern Europe and this suggests a convergent motion of the Armorican block and stable Europe.

Briden and Duff (1981) placed the pole of stable Europe during Silurian time (Fig. 6) at 330°E, 0°. For the same period, Brock determined the mean Gondwana pole to be at 30°E, 20°N whereas Crowell (1981) found a mean position of 345°E, 30°S. These last two positions for Gondwana are different but considering their mean position ($\simeq$ 20°E, 5°S), it is probable that our Vendée pole (321°E, 28°S) is more comparable to that of stable Europe suggesting that the Armorican block was located in the vicinity of stable Europe at that time.

In summary, the scenario of the motions (rotations) of the Armorican plate with respect to stable Europe and Gondwana could be as follows:

At Upper Precambrian and Lower to Middle Cambrian times, the Armorican block and Gondwana had a common APW path. They were in juxtaposition and were displaced together (Hagstrum et al., 1980). In Middle to Upper Cambrian time, the Armorican plate broke away from Gondwana but was located at a large distance from stable Europe. At Ordovician time, the Armorican block was moving in the

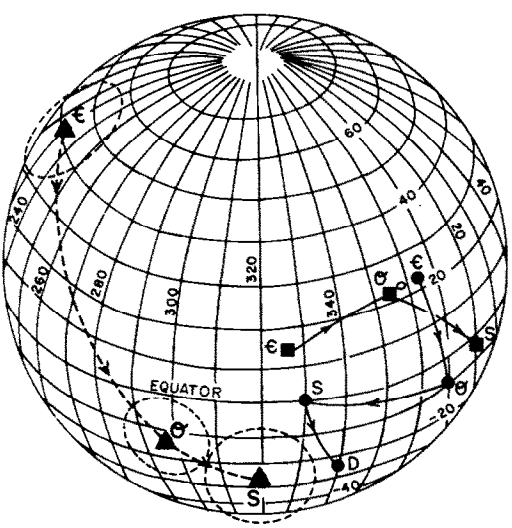


Fig. 6. Paleomagnetic pole positions for the period: Upper Cambrian–Devonian for stable Europe (circles) and the period: Upper Cambrian–Silurian for Gondwana (squares) and this study (Vendée) illustrated with triangles. $\mathcal{C}$ = Cambrian, O = Ordovician, S = Silurian and D = Devonian. The ovals representing the probable error at the 95% confidence level are shown for the palaeopoles of the Vendée (this study).

TABLE III

Palaeopoles of northwestern Europe with the exception of Soviet Union poles obtained from formations located west of the Urals

Lithological unit	Location	Palaeopole position		d_p	$\alpha_{95}(°)$	d_m	Reference
		Cambrian					
Nexø sandstone	Danemark	38°N	256°E	7		11	Prasad and Sharma (1978)
Fen Complex	Norway	63°N	142°E	3		4	Poorter (1972)
St. Quay diorite	France	38°N	262°E		12		Hagstrum et al. (1980)
Keralain gabbro	France	31°N	268°E		4		Hagstrum et al. (1980)
SE granite	Jersey Island	77°N	337°E	9		10	Duff (1978)
SW granite	Jersey Island	74°N	356°E	8		9	Duff (1978)
Volcanites	Jersey Island	52°N	323°E	33		35	Duff (1978)
St. Peter Port gabbro	Guernsey Island	13°N	224°E		11		Hailwood and Garrett (1977)
Gwna Group	Wales	17°N	237°E	–		–	Piper (1976)
Quartzite and rhyolite	France	40°N	222°E		19		this study
Mean (excluding this study), $N = 9$		52°N	251°E		25		
		Ordovician					
Eycott Group	England	07°S	357°E	4		7	Briden and Morris (1973)
Borrowdale volcanites	England	00°	023°E	7		9	Faller et al. (1977)
Builth volcanites	England	16°S	359°E	6		11	Piper and Briden (1973)

Builth intrusives	England	02°S	002°E	11		16	Piper and Briden (1975)
Mweelrea Ignimbrites	Ireland	11°S	038°E	–		–	Morris et al. (1975)
Ballantrae gabbro	Scotland	10°S	026°E	11		19	Piper (1978)
Ballantrae serpentinite	Scotland	12°S	027°E	6		10	Piper (1978)
Aberdeenshire gabbros	Scotland	11°S	009°E	6		10	Sallomy and Piper (1973a)
Portrane volcanites	Ireland	32°S	326°E	–		–	Morris (1976)
Balbriggan volcanites	Ireland	29°S	291°E	–		–	Morris (1976)
Fishguard volcanites	Wales	02°S	299°E	16		28	Morris et al. (1973)
La Meilleraie sandstone	France	15°S	293°E		14		this study
Mean (excluding this study), $N = 11$		14°S	357°E		22		
		Silurian					
Lorne volcanites	Scotland	02°N	321°E	5		8	Latham and Briden (1975)
Old Red SS. volcanites	England	04°S	320°E	4		7	Sallomy and Piper (1973b)
Ringerike sandstone	Norway	21°S	339°E	–		–	Storetvedt et al. (1968)
Llandovery traps	England	09°N	287°E	9		16	Morris et al. (1973)
Arrochar Complex	Scotland	08°S	324°E		5		Briden (1970)
Garabal Hill–Glen Fyne	Scotland	05°S	326°E		23		Briden (1970)
La Lombardière Basalts	France	28°S	321°E		22		this study
Mean (excluding this study), $N = 6$		06°S	320°E		17		

Definitions: N = number of palaeopoles used in this statistical study; d_m = short axis of the oval of 95% confidence about the pole position; d_p = long axis of the oval of 95% confidence about the pole position; α_{95} = see Table I.

direction of stable Europe and was located closer to stable Europe than to Gondwana. By Silurian time, the Armorican block was even closer to stable Europe, within 15°–20° or 1000–2000 km. By Late Devonian and Early Carboniferous time, there is no or little separation left between the Armorican Massif and stable Europe (Jones et al., 1979), but there still exists a separation between the Armorican Massif and stable Europe on one hand and Gondwana on the other.

At this stage, it is nevertheless important to point out that identical palaeopoles do not necessarily mean the unicity of the blocks considered because no constraint is put on the longitude. One segment of the blocks could very well be agglomerated in one specific position and another part could be tied up to another continent in another position. Generally speaking, Lower Palaeozoic reconstructions are rather speculative since markers are lacking. This study area is located in a region considered to represent a continental block south of the hercynian orogen (Gapais and Le Corre, 1980) and so in the eventuality of a collision, this block could have been welded to another continent.

DISCUSSION

This study partially fills the gap for the palaeomagnetic data during the period: Upper Cambrian–Silurian in the Armorican block. Even though there are definite indications that tectonic rotations took place at either regional or local scale within this block, it is desirable to calculate the palaeopoles for the Armorican block so as to enable a comparison between it and the related European and Gondwana plates as a function of time. The results pertaining to palaeolatitudes could eventually be simplified when additional palaeomagnetic studies will be available and when regional or local rotations will be taken into consideration. Considering that the palaeomagnetic data of Gondwana for the Upper Cambrian–Silurian period are limited in number and uncertain in character, the discussion pertaining to palaeolatitudes of this plate will be omitted.

During Early Cambrian time, a stable Europe was located at low latitudes and relatively near to the equator. In Early–Middle Cambrian time, northwestern Europe was displaced towards higher latitudes in the western hemisphere. In Middle to Late Cambrian time, stable Europe moved in a northerly direction towards low latitudes. In Ordovician time, the northwestern European continent was located at approximately 30°S of the equator. Between the Ordovician and the Silurian, stable Europe underwent an anticlockwise rotation around a vertical axis and in Devonian time, northwestern Europe was situated at a latitude close to the equator.

During the Late Cambrian, the Armorican block was situated at low latitudes (slightly south of the equator, $\simeq -5°$) as was the case for stable Europe. In Ordovician time, the Armorican block was located almost at the equator (3°S) and thus was separated by 30° of latitude from stable Europe. It is at this time that the Armorican block was at its closest position relative to the equator. This result is not

surprising because redbeds, such as those of the Armorican sandstone, are generally characteristic of tropical and subtropical regions. Finally, in Silurian time, the Armorican block was moving towards the southern hemisphere (6°S).

Because the palaeolatitudes of the Armorican block remained generally similar between Late Cambrian and Silurian time and because the palaeopole positions are very variable during the same geological timespan, it would appear that the Armorican block underwent several rotations about a subvertical axis in the course of its longitudinal motion from Gondwana to stable Europe during the period between Late Cambrian and Devonian–Carboniferous. To the extent that such complicating situations have existed, the reconstruction of blocks which compose present-day Europe presents a difficult task. This study suggests that there was a smaller angular difference between stable Europe and the Armorican block during the Silurian than during the Ordovician, implying that the Armorican Massif was incorporated to stable Europe probably in Devonian time. This timing coincides almost with the beginning of the Hercynian orogeny. One can then conclude that this orogeny may be related to a collision event between the Armorican block and the stable European plate. However, in the context that the Hercynian orogeny represents part of an intercontinental phenomenon related to the collision of the Gondwana supercontinent with the European plate, the Armorican block may have already been accreted to stable Europe and, having acted as a buffer zone, was imprinted by various intraplate deformations. Jones et al. (1979) concluded that the subduction zone (or collision) took place at the southern limit of the Hercynian belt. In many aspects, this Hercynian orogeny may thus have an ensialic origin. Inasmuch as the Vendean region was considered unfavourable for palaeomagnetic studies because of serious complications due to Hercynian overprinting, this first study would indicate that overprinting is not that great and that useful information can be derived from the southern Armorican block. This points to the possibility of fruitful additional palaeomagnetic research not only in the southern part of the Armorican block but also in Massif Central.

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REFERENCES

Argand, E., 1923. La tectonique de l'Asie. C.R., Congr. G2uol. Int., 13e, 1922, Liège, pp. 169–371.

Autran, A. and Cogné, J., 1980. La zone interne de l'orogène varisque dans l'Ouest de la France et sa place dans le développement de la chaîne hercynienne. In: Géologie de l'Europe. Colloq. C6, 26e Congr. Géol. Int., Paris, pp. 90–111.

Briden, J.C. and Duff, B.A., 1981. Pre-Carbonifereous paleomagnetism of Europe north of the Alpine orogenic belt. In: M.W. McElhinny and D.A. Valencio (Editors), Paleoreconstruction of the Continents. Am. Geophys. Union, Geodyn. Ser., 2: 137–150.

Briden, J.C. and Morris, W.A., 1973. Paleomagnetic studies in the British Caledonides. III. Igneous rocks of the northern Lake District, England. Geophys. J. R. Astron. Soc., 34: 27–36.

Brock, A., 1981. Paleomagnetism of Africa and Madagascar. In: M.W. McElhinny and D.A. Valencio (Editors), Paleoreconstruction of the Continents. Am. Geophys. Union, Geodyn. Ser., 2: 65–76.

Carte Géologique de la France (C.G.F.) et de la marge continentale, 1980. Réalisation du Bureau de Recherches Géologiques et Minières. Scale 1/1,500,000.

Cogne, J. and Wright, A.E., 1980. L'orogène cadomien. Vers un essai d'interprétation paléogéodynamique unitaire des phénomènes orogéniques fini-précambriens d'Europe mayenne et occidentale, et leur signification à l'origine de la croûte et du mobilisme varisque puis alpin. In: Géologie de l'Europe. Colloq., 26e Congr. Géol. Int., Paris, pp. 29–55.

Crowell, J.C., 1981. Early Paleozoic Glaciation and Gondwana Drift. In: M.W. McElhinny and D.A. Valencio (Editors), Paleoreconstruction of the Continents. Am. Geophys. Union, Geodyn. Ser., 2, pp. 45–49.

Duff, B.A., 1978. Palaeomagnetism and rock magnetic studies of Lower Palaeozoic rocks on Jersey and the adjacent regions of the Armorican Massif. Ph.D. Thesis, Univ. Leeds.

Duff, B.A., 1979. The palaeomagnetism of Cambro-Ordovician redbeds, the Erquy Spilite Series and the Trégastel–Ploumanac'h granite complex, Armorican Massif (France and the Channel Islands). Geophys. J.R. Astron. Soc., 59: 345–365.

Duff, B.A., 1980. The palaeomagnetism of Jersey volcanics and dykes, and Lower Palaeozoic apparent polar wander path for Europe. Geophys. J.R. Astron. Soc., 60: 355–375.

Durand, S. (Coordinator), et al., 1977. Bretagne; guides géologiques régionaux. Masson, Paris, 208 pp.

Faller, A.M., Briden, J.C. and Morris, W.A., 1977. Palaeomagnetic results from the Borrowdale Volcanic Group, English Lake District. Geophys. J.R. Astron. Soc., 48: 111–121.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London, Ser. A, 217: 295–305.

Gabilly, J., et al., 1978. Poitou, Vendée, Charentes; guides géologiques régionaux. Masson, Paris, 200 pp.

Gapais, D. and Le Corre, C., 1980. Is the Hercynian belt of Brittany a major shear zone? Nature, 283: 574–576.

Hagstrum, J.T., Van der Voo, R., Auvray, B. and Bonhommet, N., 1980. Eocambrian–Cambrian palaeomagnetism of the Armorican Massif, France. Geophys. J.R. Astron. Soc., 61: 489–517.

Hailwood, E.A. and Garrett, C., 1977. Palaeomagnetism of the dynamo-thermal history of the St. Peter Port Gabbro, Guernsey. Proc. Ussher Soc., 4: 26–35.

Jones, M., Van der Voo, R. and Bonhommet, N., 1979. Late Devonian to Early Carboniferous palaeomagnetic poles from the Armorican Massif, France. Geophys. J.R. Astron. Soc., 58: 287–308.

Latham, A.G. and Briden, J.C., 1975. Palaeomagnetic field directions in Siluro-Devonian lavas of the Lorne Plateau, Scotland and their regional significance. Geophys. J.R. Astron. Soc., 43: 233–242.

Mathieu, G., 1961. Un tableau stratigraphique sur le bord nord du bassin d'Aquitaine. Bull. Soc. Géol. Fr., 7e Sér., 3: 11–20.

Mathieu, G., 1972. Contribution à la connaissance des terrains anciens de la région de la Gâtine, des Mauges et du Bocage vendéen (France). Ann. Soc. Géol. Nord, 92(4): 181–198.

Mathieu, G., 1976. La roche-sur-Yon, Carte géologique de la France, 1/80,000; texte—6 p.

Matthews, S.C., Chauvel, J.J. and Robardet, M., 1980. Variscan geology of northwestern Europe. In: Géologie de l'Europe. Colloq. C6, 26e Congr. Géol. Int., pp. 69–76.

McElhinny, M.W., 1964. Statistical significance of the fold test in palaeomagnetism. Geophys. J.R. Astron. Soc., 8: 338–340.

McElhinny, M.W., 1974. Palaeomagnetism and Plate Tectonics. Cambridge University Press, London.

Morel, P. and Irving, E., 1978. Tentative paleocontinental maps of the Early Phanerozoic and Proterozoic. J. Geol., 86: 535–561.

Morel, P. and Irving, E., 1980. The concept of a mobile Pangea and the continuity of continental drift. J. Geomagn. Geoelectr., 32 (Suppl. III): S11139–S11145.

Morel, P. and Irving, E., 1981. Paleomagnetism and the Evolution of Pangea. J. Geophys. Res. 86 (B3): 1858–1872.

Morris, W.A., 1976. Paleomagnetic results from the Lower Paleozoic of Ireland. Can. J. Earth Sci., 13: 294–304.

Morris, W.A., Briden, J.C., Piper, J.D.A. and Sallomy, J.T., 1973. Palaeomagnetic studies in the British Caledonides. V. Miscellaneous new data. Geophys. J.R. Astron. Soc., 34: 69–105.

Piper, J.D.A., 1976. Magnetic properties of Precambrian pillow lavas of the Mona Complex and related dyke swarm, Anglesey, Wales. Geol. J., 11: 189–201.

Piper, J.D.A., 1978. Palaeomagnetism and palaeogeography of the southern Uplands block in Ordovician times. Scott, J. Geol., 14: pp. 93–107.

Piper, J.D.A. and Briden, J.C., 1973. Paleomagnetic studies in the British Caledonides. I. Igneous rocks of the Builth Wells–Llandridnod Wells Ordovician inlier, Radnoshire, Wales. Geophys. J.R. Astron. Soc., 34: 1–12.

Poorter, R.P.E., 1972. Preliminary palaeomagnetic results from the Fen Carbonatite complex, South Norway. Earth Planet. Sci. Lett., 17: 194–198.

Prasad, S.N. and Sharma, P.V., 1978. Palaeomagnetism of the Nexø Sandstone from Bornholm Island, Denmark. Geophys. J.R. Astron. Soc., 54: 669–680.

Sallomy, J.T. and Piper, J.D.A., 1973a. Palaeomagnetic studies in the British Calcedonides. II. The Younger Gabbros of Aberdeenshire. Geophys. J.R. Astron. Soc., 34: 13–76.

Sallomy, J.T. and Piper, J.D.A., 1973b. Palaeomagnetic studies in the British Caledonides. IV. Lower Devonian Lavas of the Strathmore region, Scotland. Geophys. J.R. Astron. Soc., 34: 47–68.

Seguin, M.K., 1975. Conception et réalisation d'un appareil de désaimantation par champs alternatifs et son utilisation dans les études paléomagnétiques. Abh. Braunschw. Wiss. Ges., 25: 25–43.

Storetvedt, K.M., Halvorsen, E. and Gjellestad, B., 1968. Thermal analysis of the natural remanent magnetism of some Upper Silurian red sandstones in the Oslo region. Tectonophysics, 5: 413–426.

Ters, M., 1972. Palluau et Ile d'Yeu, Carte géologique de la France, 1/80,000; texte—60 pp.

Van der Voo, R. and Klootwijk, C.T., 1972. In: E. Irving, E. Tanczyk and J. Hastie (Editors), Catalogue of Paleomagnetic Directions and Poles; Paleozoic Results 1949–1975. Geomagn. Ser., 2, 1976, Ottawa.

Van der Voo, R., Briden, J.C. and Duff, B.A., 1980. Late Precambrian and Paleozoic paleomagnetism of the Atlantic-bordering continents. In: Géologie de l'Europe. Colloq. C6, 26e Congr. Géol. Int., Centre National de la Recherche Scientifique, Paris, pp. 203–212.

Wilson, J.T., 1966. Did the Atlantic close and then re-open? Nature, 211: 676–681.

Wyns, R., 1979. Découverte de Graptolites dans les séricitoschistes associés aux phtanites du synclinorium de Chantonnay (Vendée) (ex-Briovérien): conséquences stratigraphiques et structurales. C.R. Acad. Sci, 288, Sér. D: 477–480.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: D.W. Collinson, K.M. Creer and S.K. Runcorn (Editors), Methods in paleomagnetism. Elsevier, Amsterdam, pp. 254–286.